

Question

Write a Java program to read a student's name, roll number, and three marks. Display the student's details, total marks, and percentage.

Steps for Execution

1. Open a Java editor or Command Prompt.
2. Save the code as `Student.java` inside the `basics` folder.
3. Compile the program using the command shown below.
4. Run the program and enter the requested details and marks.

Algorithm

1. Start the program.
2. Read the student's name, roll number, and three marks.
3. Add the marks to calculate the total.
4. Calculate the percentage by dividing the total by three.
5. Display the student's details and results.
6. Stop the program.

UML Class Diagram

```
+-----+
|           Student           |
+-----+
| + name: String              |
| # RollNo: int               |
| # marks: int[]              |
+-----+
| + Student(name, RollNo, marks) |
| + getTotalMarks()            |
| + percentage()               |
| + displayDetails()           |
| + main(args)                 |
+-----+
```

CODE: SRC

```
package basics;
import java.util.Scanner;

public class Student {
    public String name;
    protected int RollNo;
    protected int[] marks;

    public Student(String name, int RollNo, int[] marks) {
        this.name = name;
        this.RollNo = RollNo;
        this.marks = marks;
    }

    int getTotalMarks() {
        int total = 0;
        for (int i : marks) {
            total += i;
        }
        return total;
    }

    void percentage() {
        int total = getTotalMarks();
        double percent = (double) total / marks.length;
        System.out.println("Percentage: " + percent);
    }

    void displayDetails() {
        System.out.println("Name: " + name.toUpperCase());
        System.out.println("Roll Number: " + RollNo);
        System.out.println("Marks: ");
        for (int mark : marks) {
            System.out.print(mark + " ");
        }
        System.out.println();
        int total = getTotalMarks();
        System.out.println("Total Marks: " + total);
        percentage();
    }
}
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter student name: ");
    String name = sc.next();

    System.out.print("Enter student roll number: ");
    int rollNo = sc.nextInt();

    System.out.print("Enter student marks: ");
    int[] marks = new int[3];
    for (int i = 0; i < 3; i++) {
        marks[i] = sc.nextInt();
    }

    Student s = new Student(name, rollNo, marks);
    s.displayDetails();

    sc.close();
}
}

```

Compilation and Execution

```

javac -d . basics/Student.java
java basics.Student

```

Output

```

Enter student name: Alice
Enter student roll number: 101
Enter student marks: 80 90 85
Name: ALICE
Roll Number: 101
Marks:
80 90 85
Total Marks: 255
Percentage: 85.0

```